AQUASHIELD

Your complete project, explained from the tiniest basics to the technical details

Urban flood-intelligence prototype for Vasai-Virar-Nalasopara (VVN), Maharashtra

WEATHER	RISK	DECIDE	RESPOND
Rain + terrain	Maps + analytics	ML + routes	Sensors + reports

One honest sentence

AquaShield combines historical weather and map-derived information in one browser app, estimates flood susceptibility with a transparent formula, demonstrates an ML model and risk-aware route preview, and keeps simulated sensors and community media separate from official facts.

Use the green-style summary boxes for what to say. The remaining sections explain what each screen and score really means.

Project snapshot: 21,912 processed place-day rows, 12 rainfall sample locations, 18,472 road segments, 65 YouTube entries and 6 manually catalogued Instagram entries.

1. Start with the tiny basics

These words explain the whole project.

Word	Tiny meaning	In AquaShield
Dataset	An organised table of examples.	Place-and-day measurements/features.
Feature	A clue supplied to a model.	Rain, elevation, drainage distance, building count and derived rain history.
Label	The example answer the model practises.	Low, Medium or High formula-generated susceptibility.
Model	A program that learns patterns.	Random Forest classifier and a score regressor.
Dashboard	Useful screens in one app.	Nine pages in the left menu.
Simulation	Computer-made example values.	IoT readings; no physical sensor is attached.

What is it?

A Python + Streamlit browser application. On this computer it opens at http://localhost:8501. It is not an Android APK and is not deployed as a cloud service.

What does dummy data mean?

Not every file is invented. Historical weather and map-derived features are used; flood classes are made by a formula; IoT and example social posts are simulated or manually catalogued; real community reports exist only if a person submits them.

AquaShield brings rainfall, terrain, drainage, roads, a model demo, route preview and future sensor/report ideas into one place. It helps explain a system design; it is not an emergency warning authority.

2. Data: what is real, derived and generated?

The strongest presenter clearly labels each layer.

Layer	What is in the project	Honest description
Rainfall	21,912 daily grid rows, 12 sample points, 2019-01-01 to 2023-12-31.	Collector code uses Open-Meteo Archive. Historical weather, not flood depth.
Elevation	Grid heights matched to rainfall locations.	Open-Meteo elevation collector; terrain feature.
Drainage	Distance to mapped water/drainage features.	Map-derived proximity proxy, not drainage capacity.
Buildings	Building counts per cell.	OSM building proxy, not a resident census. Not used in the label formula.
Roads	18,472 segments with infrastructure scores.	OSM-derived geometry/features, not live road water.
Known flood events	4 entries for 1 event date(s).	One major 18 July 2021 event; too little to validate general prediction.

Historical flood labels are constructed from rainfall, elevation and distance to mapped drainage/water. The formula uses a weighted score: rainfall 60%, elevation 25%, drainage proximity 15%. Low is below 30; Medium is 30 to below 60; High is 60 or above.

Score part	Plain meaning
Rainfall	More daily rain raises the score; capped at 100. Formula: min(rain_mm / 60, 2) x 50.
Elevation	Lower locations score higher. Formula: max(0, (20 - elevation_m) / 20) x 100.
Drainage proximity	Within 2 km raises this score in this prototype. This is a simplifying, debatable assumption.

Key distinction

Weather values can be historical observations. The Low/Medium/High target is still a formula label, not a surveyed flood/no-flood answer.

3. Walk through the app pages

The left sidebar contains nine sections.

Page	What appears	What to say
Home Dashboard	Rain summary, high/medium sample counts, forecast bars, curated events, latest grid table.	Forecast colour is a simple rain threshold; historical labels come from a weighted formula.
Flood Risk Map	12 sample points, coloured circles, interpolated heat layer, optional video pins.	Heat is a visual interpolation, not a measured boundary. Social pins are unverified.
Analytics	Rainfall line, risk-class pie, average rain by calendar month.	These charts describe weather and formula labels, not a count of real floods.
ML Model	Validation metrics and manual what-if sliders.	The model predicts proxy labels; it does not auto-issue a forecast warning.
Road Risk Map	Sampled road midpoints coloured by infrastructure score.	Static susceptibility, not current water on a road.

Analytics graph meanings

- Rainfall trend: x-axis is date; y-axis is mean rain over the sample grid in millimetres/day. Dotted marker is the July 2021 event date.
- Risk pie counts all 21,912 processed formula labels: Low 11,912 (54.4%), Medium 9,703 (44.3%), High 297 (1.4%).

- Monthly rainfall bars pool locations and years to show seasonality; this hides year-to-year differences.

4. ML model: how it learns

ML means machine learning: learn a pattern from examples, then apply it to inputs.

The dataset is split to test the model on data it did not train on. Candidate classifiers are Logistic Regression, Random Forest and XGBoost when available. Random Forest is the saved best classifier. A separate Random Forest regressor predicts the formula score.

Input feature	Easy meaning
precipitation_mm	Daily rainfall.
elevation_m	Ground height.
building_count	Nearby building proxy; included in ML but not the label formula.
drainage_dist_km	Distance to mapped water/drainage.
rain_3d, rain_7d	Rolling rain sums over recent 3 and 7 days.
rain_delta	Rainfall change from the previous day.
is_monsoon	1 for June-September; otherwise 0.
rain_elev_interact	Rainfall divided by elevation plus one.

Training steps

- Engineer rolling rainfall and season features; remove incomplete rows.
- Temporal hold-out: train on 2019-2022, evaluate on 2023.
- Spatial hold-out: set aside 3 of the 12 locations and test on those locations.
- SMOTE oversamples rare classes in temporal training data only; never the test rows.
- Choose the classifier using 60% Macro-F1 plus 40% High-class recall, not accuracy alone.
- Final saved model is fit on all rows for manual what-if inference.

What the sliders do

You choose rainfall, elevation, drainage distance, building count and monsoon status, then click Run model. This is a manual example, not an automatic live prediction.

5. ML performance: what the numbers mean

Say the validation type and the proxy-label limitation together.

Check	Data held aside	Accuracy	Macro-F1	High recall
Temporal	2023 dates; 4,380 rows	99.91%	0.991	100.0%
Spatial	3 unseen points; 5,478 rows	91.44%	0.914	96.1%

Spatial per-class recall: Low 100.0%, Medium 84.2%, High 96.1%. The spatial test contains 77 High examples. The temporal test contains 40 High examples.

Metric glossary

Metric	Tiny explanation
Accuracy	Fraction of all test rows classified correctly; common classes can dominate.
Macro-F1	Average class F1 with equal importance for Low, Medium and High.
High recall	Of High-label rows, how many did the model find? It does not tell false-alarm rate.
RMSE / MAE	Regressor error versus formula score. Stored RF values: RMSE 0.159 and MAE 0.026. Not error in centimetres of floodwater.

Honest interpretation

Near-perfect temporal scores happen because the model is learning labels created by a formula from related features. The spatial check tests unseen grid points, but their labels come from the same formula. These are genuine measured scores for reproducing proxy labels, not measured accuracy on real floods.

6. Roads and Safe Routes

A route demo is not emergency navigation.

Road Risk Map

- The map draws up to 2,800 road-segment midpoints for speed.
- Infrastructure score combines 70% low-elevation score and 30% drainage-proximity score.
- High is static proxy susceptibility; it does not mean the road is currently flooded.

Safe Routes

- You select a start and destination. Each is snapped to a nearby road midpoint.
- Dijkstra searches for the lowest sum of link weights. The app compares distance-only and risk-weighted paths.
- A risk weight uses distance multiplied by an infrastructure-risk factor.
- Only recent verified user-submitted reports can add a local route penalty; synthetic demo posts and unverified reports are ignored.

Important limitation

The graph connects nearby road midpoints with straight-line links. It is not verified street topology and does not provide turn-by-turn navigation. Check real road conditions before travelling.

7. IoT, community and social pages

These demonstrate how live information could fit into a future system.

IoT Sensor Monitoring

- Shows generated water level (cm), rain rate (mm/hour), soil moisture (%) and temperature (C).
- Water chart has example warning at 80 cm and alert at 120 cm.

Say this

These readings are simulated. No ESP32 sensor or Firebase connection is attached.

Community Reports

- A person can submit a caption, optional location and photo/video. User uploads start Unverified.
- SQLite database and media files are local to the computer; there is no shared cloud store or secure production moderation.
- Current database: 4 synthetic demo posts and 0 user uploads. One synthetic sample is marked Verified to demonstrate UI status; it is still synthetic and is ignored by route scoring.

Social Videos

- 65 public YouTube records and 6 manually catalogued Instagram entries. Instagram has no search/API integration here.
- Filters select platform, severity and locality; map pins are approximate and inferred from title/hashtags.
- Keyword-inferred severity and an unverified clip do not prove a location is flooded.

Keep these layers separate

Official/curated event context, user uploads, synthetic posts, simulated sensors and social videos are different evidence types.

8. How the pieces fit and how to run it

The app code and the training/data code have separate jobs.

Part	Main files	Job
Interface	src/dashboard/app.py, src/dashboard/maps.py	Build the nine Streamlit pages, charts and maps.
Data	data/raw, data/processed	Inputs and joined feature tables.
Preparation	src/data_pipeline/preprocess_v2.py	Join rainfall with terrain/map proxies and generate formula labels.
ML	src/model_training/train_flood_risk_model.py, models/	Train/evaluate candidates and save model plus metrics.
Routing	src/routing/graph.py	Build an approximate graph and compute paths.
IoT	src/iot/simulator.py	Generate simulated readings.
Community/social	src/community, src/social, data/social	Local uploads and distinct media catalogues.

Start the application

1. Extract the ZIP to a normal folder. 2. Install Python 3.10 or newer if needed. 3. Double-click start_aquashield.bat. 4. On first run, allow package setup to finish; internet is needed. 5. Open http://localhost:8501. 6. Leave the terminal window open while presenting.

Included in the share ZIP

The app source, data CSV/JSON, saved classifier and regressor, evaluation metrics, requirements, setup instructions and this project guide. The local virtual environment, Git internals, saved community database and uploaded media are excluded.

9. Presentation script, Q&A; and glossary

You do not need to memorise the entire guide.

90-second introduction

AquaShield is an urban flood-intelligence prototype for Vasai-Virar-Nalasopara. It combines historical rainfall and map-derived terrain, drainage and road features in one browser application. A transparent formula assigns Low, Medium or High susceptibility, and a Random Forest learns to reproduce those labels. The app also demonstrates risk maps, analytics, a route-risk preview, simulated IoT readings and separate community/social layers. The evaluation scores measure agreement with formula labels, not proven real-world flood prediction.

Likely questions

Question	Short answer
Why Macro-F1?	High is rare, so Macro-F1 gives each class equal weight.
Is accuracy 99.9% real flood accuracy?	No. It is accuracy against formula labels in a held-out time period.
Are sensors live?	No, their values are simulated; hardware integration is future work.
Instagram automatic?	No. Six manually catalogued entries; no Instagram search API.
Are routes guaranteed safe?	No. Approximate graph demonstration, not turn-by-turn navigation.
What should happen next?	Collect multiple verified flood/no-flood events and use true connected roads and road closures.

Glossary

VVN = Vasai-Virar-Nalasopara. OSM = OpenStreetMap. Dijkstra = shortest-path search on weighted links. Hold-out = data saved for testing. Proxy label = a stand-in answer generated by a rule instead of direct observation.

Reference note: the available AquaShield introduction/abstract describes the VVN focus and intended rainfall, elevation, drainage, road, ML, route and future-IoT integration. The specifically named architecture PDF was not found in the supplied folders. This guide was checked against the current app, datasets and saved metrics.

Final reminder

Explain what the prototype demonstrates and what evidence it still needs. Clear limits make the presentation credible.